

Ordering Fractions and Mixed Numbers

Answer Key

Grade 4–5

Quick Answers

- | | |
|---|---|
| 1. $>$ | 8. $\frac{5}{12}, 1 \cdot \frac{1}{2}, \frac{4}{6}$ |
| 2. $1 + \frac{3}{4}$ | 9. (a) $2 \frac{1}{4}$ written in fourths = $\frac{9}{4}$, (b) comparison symbol = $=$, $-$ |
| 3. $1 \cdot \frac{1}{2}, \frac{2}{3}, \frac{5}{6}$ | 10. $\frac{8}{12}, \frac{7}{9}, \frac{5}{6}$ |
| 4. $\frac{3}{8}$ against one half = $<$, $\frac{5}{9}$ against one half = $>$, final comparison = $<$ | 11. $1 + 1 \cdot \frac{1}{4}, \frac{3}{2}, \frac{13}{8}$ |
| 5. $\frac{7}{3}$ | 12. (a) ribbon A against ribbon B = $=$, (b) ribbon C against ribbon B = $<$, $-$ |
| 6. $\frac{3}{4}, \frac{5}{6}, \frac{7}{8}$ | |
| 7. $1 + 1 \cdot \frac{1}{2}, 1 + \frac{3}{4}, \frac{15}{8}$ | |

What is verified

10 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 9, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 4–5

4.NF.A – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–5 of 5 across 18 tagged checks.

- Write the correct symbol: $\frac{5}{8} \text{ -- } \frac{1}{2}$
Solution: One half in eighths is $\frac{4}{8}$, and 5 eighths is more than 4 eighths, so $\frac{5}{8} \boxed{>} \frac{1}{2}$
- Write $\frac{7}{4}$ as a mixed number.
Solution: $7 \div 4 = 1$ remainder 3, so 7 fourths is 1 whole and 3 fourths: $1\frac{3}{4}$
- Order least to greatest in sixths: $\frac{2}{3}, \frac{1}{2}, \frac{5}{6}$
Solution: In sixths: $\frac{2}{3} = \frac{4}{6}$, $\frac{1}{2} = \frac{3}{6}$, and $\frac{5}{6}$ stays. Comparing 3, 4, and 5 sixths gives $\frac{1}{2} \boxed{<} \frac{2}{3} \boxed{<} \frac{5}{6}$
- Use the benchmark $\frac{1}{2}$ to compare $\frac{3}{8}$ and $\frac{5}{9}$.
Solution: Half of 8 is 4, and $3 < 4$, so $\frac{3}{8} \boxed{<} \frac{1}{2}$. Half of 9 is 4.5, and $5 > 4.5$, so $\frac{5}{9} \boxed{>} \frac{1}{2}$. One fraction is below the benchmark and the other above it, so $\frac{3}{8} \boxed{<} \frac{5}{9}$
- Point P sits at $2\frac{1}{3}$. Write P as an improper fraction in thirds.

Solution: Two wholes hold $2 \times 3 = 6$ thirds, plus 1 more third makes 7 thirds: $\boxed{\frac{7}{3}}$

6. Order least to greatest: $\frac{7}{8}, \frac{3}{4}, \frac{5}{6}$

Solution: In twenty-fourths: $\frac{7}{8} = \frac{21}{24}, \frac{3}{4} = \frac{18}{24}, \frac{5}{6} = \frac{20}{24}$. Ordering 18, 20, 21 gives $\boxed{\frac{3}{4}} < \boxed{\frac{5}{6}} < \boxed{\frac{7}{8}}$

7. Order least to greatest as eighths: $1\frac{3}{4}, \frac{15}{8}, 1\frac{1}{2}$

Solution: In eighths: $1\frac{3}{4} = \frac{14}{8}, \frac{15}{8}$ stays, $1\frac{1}{2} = \frac{12}{8}$. Ordering 12, 14, 15 eighths gives $\boxed{1\frac{1}{2}} < \boxed{1\frac{3}{4}} < \boxed{\frac{15}{8}}$

8. Trails: Pine $\frac{5}{12}$ mi, Oak $\frac{4}{6}$ mi, Elm $\frac{1}{2}$ mi. Order least to greatest.

Solution: Benchmark check: half of 12 is 6, and $5 < 6$, so Pine is under $\frac{1}{2}$. Half of 6 is 3, and $4 > 3$, so Oak is over $\frac{1}{2}$. Elm is exactly the benchmark. So $\boxed{\frac{5}{12}} < \boxed{\frac{1}{2}} < \boxed{\frac{4}{6}}$

9. Ben says $2\frac{1}{4} < \frac{9}{4}$ because 2 is less than 9. (a) Write $2\frac{1}{4}$ in fourths. (b) Correct symbol. (c) Explain.

Solution: (a) Two wholes are 8 fourths, plus 1 fourth: $2\frac{1}{4} = \boxed{\frac{9}{4}}$

(b) Both names stand for 9 fourths, so $2\frac{1}{4} \boxed{=} \frac{9}{4}$

(c) (*instructor judges*) Ben compared the whole number 2 with the numerator 9 — two different kinds of numbers. Rewriting $\frac{9}{4}$ shows it is 2 wholes and 1 fourth, the very same value, so neither is smaller.

10. Order least to greatest: $\frac{5}{6}, \frac{7}{9}, \frac{8}{12}$

Solution: In thirty-sixths: $\frac{5}{6} = \frac{30}{36}, \frac{7}{9} = \frac{28}{36}, \frac{8}{12} = \frac{24}{36}$. Ordering 24, 28, 30 gives $\boxed{\frac{8}{12}} < \boxed{\frac{7}{9}} < \boxed{\frac{5}{6}}$

11. Order least to greatest using benchmarks 1 and $1\frac{1}{2}$: $\frac{13}{8}, 1\frac{1}{4}, \frac{3}{2}$

Solution: All three are between 1 and 2. $1\frac{1}{4}$ is below $1\frac{1}{2}$; $\frac{3}{2}$ is $1\frac{1}{2}$; and $\frac{13}{8} = 1\frac{5}{8}$ is above it. So $\boxed{1\frac{1}{4}} < \boxed{\frac{3}{2}} < \boxed{\frac{13}{8}}$

12. Ribbons: A is $1\frac{5}{6}$ m, B is $\frac{11}{6}$ m, C is $1\frac{2}{3}$ m. (a) Compare A and B. (b) Compare C and B. (c) Shortest?

Solution: (a) In sixths, $1\frac{5}{6} = \frac{11}{6}$ — the same number, so $1\frac{5}{6} \boxed{=} \frac{11}{6}$

(b) $1\frac{2}{3} = \frac{10}{6}$, and 10 sixths is less than 11 sixths, so $1\frac{2}{3} \boxed{<} \frac{11}{6}$

(c) (*instructor judges*) Ribbon C is the shortest. Ribbons A and B tie because 11 sixths regroups as 1 whole and 5 sixths — two names for one length.